

GenAI in the classroom: Customized GPT roleplay for process safety education

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ABSTRACT

This paper outlines innovative pedagogical approaches in chemical engineering safety education, utilising three key strategies: (1) Integration of the direct experiences of experienced professional engineers to teach expertise-aligned process safety case studies; (2) Shifting from traditional short oral presentations to more comprehensive and dynamic ‘engineering meeting’ formats to drive constructive student learning; and (3) The use of Generative AI to enhance (1) and (2), with cost/time scalability, improved student access and by accommodating greater learning-style diversity. Evaluation of these educational innovations is performed through a mixed-methods approach and reveals positive impacts on student learning and engagement. The paper provides a detailed outline of classroom implementation, with supporting resources, for straightforward integration by other academics.

1. Introduction

1.1. Process safety education

The effective and meaningful teaching of process safety principles has become a primary focus within process engineering and chemical engineering education (Pollock and Sorensen, 2018; Amaya-Gómez et al., 2019; Fogler and Hirshfield, 2021; Pollock and Sorensen, 2021; Qian et al., 2023). Teaching process safety principles is considered essential, primarily because it's seen to reduce the likelihood of future incidents to create safer workplaces, improve efficiency and reduce risk (to capital, environment and personnel). It also provides an engaging format to segue into a range of complimentary learning objectives into the classroom, such as a technical understanding of the safety issues themselves (Qian et al., 2023) and the broader ethical discussions around professional practice, environmental risk mitigation and public safety obligations (Guntzburger et al., 2017; Martin et al., 2021; Sandri et al., 2023).

Process safety discussions can be integrated into a chemical engineering curriculum in a wide range of ways, from a macroscale or strategic level, by integrating safety discussions throughout the curriculum as a matter of policy or through coherent curriculum design

(Perrin et al., 2018; Pollock and Sorensen, 2018). Process Safety, Hazard Awareness and Personal Safety discussions may also be implemented at the microscale or tactical level, for example by considering specific learning activities within individual subjects. At this subject-specific level, there are a broad range of tactics and learning designs that can be employed to convey safety concepts (Hassall et al., 2020), for example Shallcross (2014) has presented safety concepts to students through *safety shares*, or short 3–5 min case studies, presented by the lecturer, at the start of every class. This approach is designed to mirror a *safety moment* in engineering meetings but is designed to always be delivered by the lecturer (rather than rotating through all participants, like a safety moment meeting). Leveneur et al. (2016) have blended safety concepts with specific technical content to offer real-world relevance to theory-based problems and Zeng et al. have developed specific training modules for engineers (Zeng and Zeng, 2017). These are all innovative and interesting formats for introducing safety concepts into chemical engineering subjects, but the most widely used pedagogical design remains the use of safety-incident case studies (typically case studies of major incidents) (Shallcross, 2013a, 2013b; García-Fayos et al., 2016; Kerin and Pollock, 2019; Dunn et al., 2020; Hassell, 2021). Here students are typically introduced to a widely known industrial incident, that had catastrophic consequences and often resulted in the

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loss of life. They are then required to complete background research on the topic, with a specific focus on the underlying causes, the context and the specific root cause safety failings, before presenting the learnings from the incident. Safety case studies are typically then assessed in short oral presentations from students (Shallcross, 2013a, 2013b; Hassell, 2021). This format is often encouraged because it mirrors the industry practice of the ‘safety moment’ (and unlike a *safety share*, it rotates through presenters). The short timeframe of the oral presentation also allows for assessment at a large scale (it is not too onerous for an academic coordinator to listen to all the student presentations).

When designing effective process safety education, it is important to be mindful of not just how it is *taught*, but also how it is *assessed* (Hassan, 2011). Assessment has been inextricably linked to the learning strategies employed by students and so the assessment itself shapes the learning behaviours the students may adopt (Hargreaves, 1997). A problem with short oral presentations is the format of the assessment itself often leads to reductive student explanations of the root cause of the safety incident. In short oral presentations, students are not given the time or space to expand into the underlying safety issues, and so by virtue of the assessment design itself, they are required to condense, ignore and over-simplify important nuances within the safety incident case study. Rather than being able to demonstrate a holistic understanding of process safety, for example a layered ‘swiss cheese’ model commonly used to understand safety root causes (Ahmad and Pontiggiaa, 2015; Shabani et al., 2023), short oral presentations drive singular cause-and-effect explanations.

An alternative long-form format for discussing and assessing students’ understanding of safety case studies is a viva (also called defence), but that we referred to as an ‘engineering meeting’, to mirror workplace experiences for more authentic assessment. In this format, small student teams (3–4 students) attend a 30 min meeting, with an academic and an industry consultant. Here they are asked questions by the 2-person panel, to prompt a discussion of their underlying understanding of the case study. The viva format allows for a more reactive and dynamic exchange between examiners and students (compared to an oral presentation, that delivers a fixed slide deck) and also mirrors a more common workplace experience (oral updates or discussions, rather than a fixed presentation).

When considering long-form oral assessment, a tension arises between the pedagogical goals (allowing students to fully explore the safety case study) and the realities of practical implementation (the time and resources required to implement such activities).

A similar, second key tension arises in how the safety material is taught. Recent literature has emphasised that good teaching practice in engineering safety is heavily contingent on the direct safety experience and qualification of the teaching staff themselves (Kerin and Pollock, 2019; Pollock and Sorensen, 2021). This suggests that engineering safety is best taught within an integrated engineering programme (Pollock and Sorensen, 2021) and can be significantly enhanced using industry consultants with direct experience in safety. In our own teaching practice, we have found an often under-utilized reservoir of experience and goodwill among the community of late-career and recently retired chemical engineers. This demographic, possessing decades of practical experience, represents a significant resource of intellectual capital, available to universities, that if managed and harnessed effectively can significantly improve safety education for the incoming generation of chemical engineers. The key challenge becomes the available time these senior engineers can dedicate to teaching students and the resources required to pay them to do so. Again, there is a tension between the need for economies of scale in content delivery with an optimized learning design, that allows individual students to interact directly with a highly qualified industry expert. In the current paper, we hope to address these competing goals through the use of GPTs within the learning design.

1.2. GenAI: GPT-based teaching tools

Since November 2022, following the general release of OpenAI’s ChatGPT-3, there has been an explosion of interest in the use of generative artificial intelligence (GenAI), with a wide range of specific use cases in tertiary education (Adeshola and Adepoju, 2023; Jacobsen and Weber, 2023; Mosaiyebzadeh et al., 2023; Qadir, 2023; Rahman and Watanobe, 2023). The literature on GPT use has been growing at a rate that is almost beyond the capacity of any single individual to keep abreast of, but ironically it has also been hypothesized that GPTs will soon provide broad literature reviews for academics (Rahman et al., 2023a, 2023b).

To date, most education-focused literature has focused on possible opportunities and risks associated with GenAI, particularly in the domain of academic integrity (Adeshola and Adepoju, 2023; Jacobsen and Weber, 2023; Preiksaitis and Rose, 2023; Qadir, 2023; Rahman and Watanobe, 2023). However there are now also complimentary studies using GPTs as academic tools, and testing their efficacy (Wandelt et al., 2023). GPT tools can provide a range of complimentary administrative functionality, for example by helping to develop subject learning objectives (Sridhar et al., 2023) or for curriculum mapping like identifying sustainable development content within curriculum for accreditation (Bury et al., 2022). Perhaps most exciting is the capacity for GPTs to offer direct instruction and teaching tools to learners. Khan Academy has released Khanmigo (an AI tutor) and OpenAI has now paired with Arizona State University for the first formal partnership to develop tutorial and discipline-specific teaching applications. Individual academics have used ChatGPT itself for a range of teaching activities (Lo, 2023; Tsai et al., 2023). In an ironic inversion, there have also been custom GPT-chatbots built to simulate students, to help train human teachers (Markel et al., 2023).

The promise of these AI-teaching tools include: improved accessibility to education, infinite scalability (particularly relevant for the on-line education market), instantaneous responses, adaptive learning (that conforms to the ability and preferences of the learner directly) and also the potential for significantly lower cost (Adeshola and Adepoju, 2023; Baidoo-Anu and Ansah, 2023; Jacobsen and Weber, 2023).

Although there are now a range of AI-tutors available (and other abstracted teaching supports) a key problem with AI-tutors are the frequency of ‘hallucinations’ (incorrect or nonsensical replies that do not align with factual reality, typically due to limitations in the model’s training dataset or due to the misinterpretation of the user input prompt) (Sovrano et al., 2023).

When designing GPTs to function as tutors, an interactive text-book or more abstracted learning assistants, there are 2 common learner expectations that implicitly accompany the tool, that mirror their expectations of all certified course material (Honig et al., 2023):

1. Reliability (performance expectancy). Learners expect the tool to be a reliable source of truth, at least comparable to a textbook or subject resources (like notes and lecture slides) where errors are rare and will be fixed if found. This level of reliability isn’t possible with current GPT models.
2. Compellability. Learners expect that the tool will always provide an answer to questions within the course scope. It’s perceived as an ‘always available’ tutor, who will reply in a form equivalent to a tutor.

This creates an expectation that the GPT must *always* provide an answer that is *correct* (Honig et al., in review). Even if the question is outside of the training dataset of the GPT, or the question is ambiguous, it must provide a best approximation. This is a much higher standard than we place on humans (teaching staff or industry professionals) who can reply ‘I don’t know’ or ‘I don’t understand your question’ or might just make mistakes sometimes, which can be accepted.

These issues of hallucinations can be mitigated by having the chatbot

roleplay human activities, as opposed framing classroom use-cases of GPTs as a search tool or conduit to factual information. For example, GPTs have been used to create simulated patient encounters for medical student diagnosis (Sardesai et al., 2024). Here there is no expectation the information provided by the ‘simulated patient’ is always ‘correct’ (patients frequently make errors). Instead the information is meant to be assessed and cross-checked by the student-doctor, to guide appropriate diagnostic tests.

In a similar learning design to ‘GPT-patients’, in the current paper we use GPTs to roleplay as industry consultants and academics (conducting a viva) and hope to structure the activities to resolve issues of hallucinations. It is important to emphasize that these GPT teaching tools offer significantly lower cost over-heads in teaching, as well improved

accessibility for students and infinite scalability for universities, so even if their learning utility can only match current teaching practice, that still offers significant advantages.

Although GPT-tools have become widespread within tertiary education, they have been more limited in the specific area of chemical engineering education, with greater focus on developing course problems (Tsai et al., 2023) and speculating about possible futures (Xuan and Daniel, 2023). However, we note that Large Language Models (LLMs) have shown promise in developing HAZOP reports (Feng et al., 2021; Ekramipoooya et al., 2023) and for assisting in hazard analysis (Diemert and Weber, 2023).

We believe that there are a range of near-term possible applications for GPTs within chemical engineering and process safety, including: in

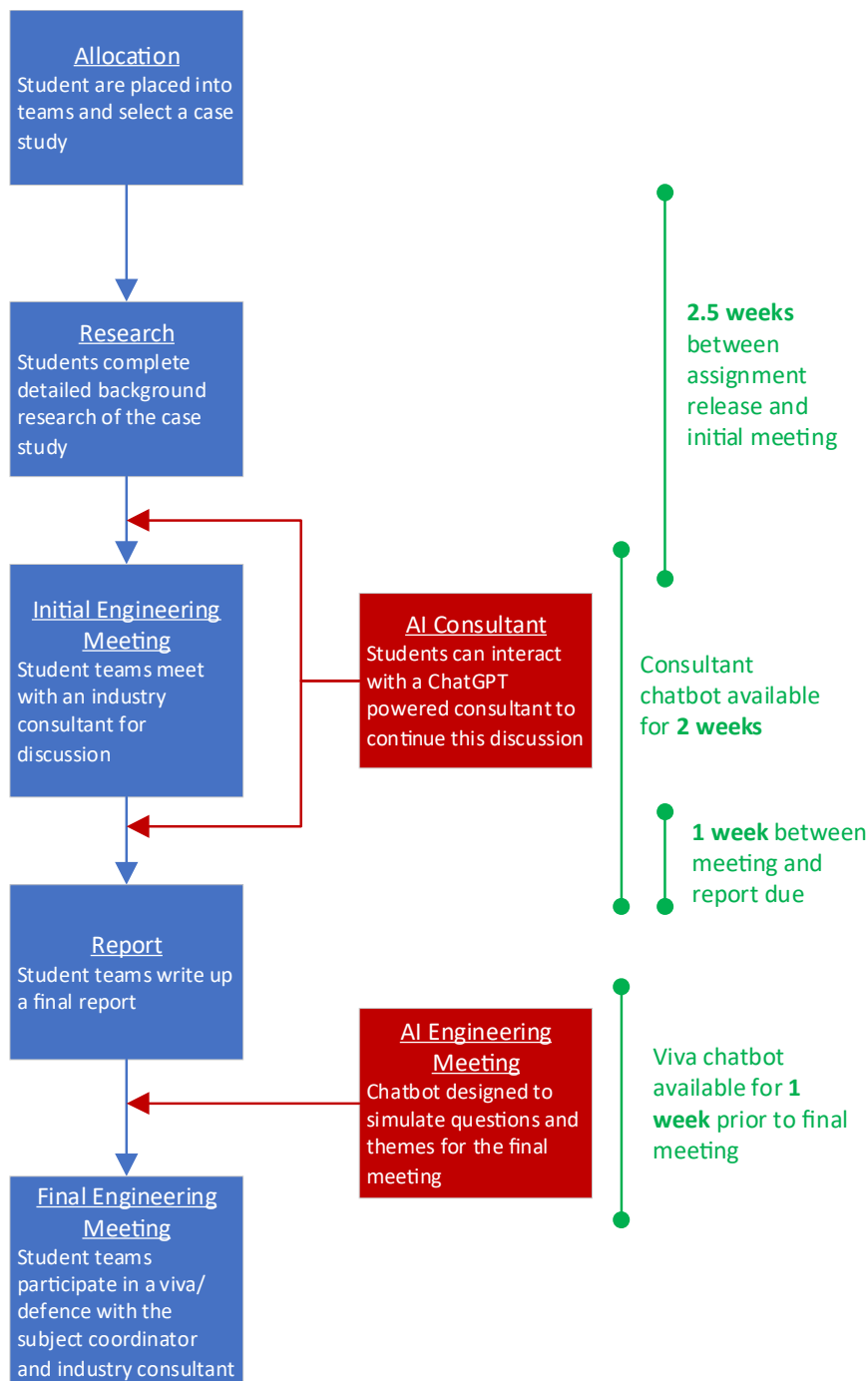


Fig. 1. Sequence of activity for the case study including opportunities to utilize GPT-based tools, with timelines.

assisting HAZOPs (proof-reading and suggesting/cross-check causes for guide words); completing and cross-checking P&IDs (using the symbolic lexicon of a P&ID schematics, rather than written languages); for safety incident investigations (for offering systematic and broader idea sets to understand possible causes). While GPT-tools are often discussed in teaching contexts, they also have direct process safety applications, that continue to be developed.

2. Design and implementation of the intervention

One of the authors (Christopher Honig) is the subject coordinator of a 3rd year undergraduate subject called ‘Safety and Sustainability Case Studies’ at the University of Melbourne, which functions as a capstone subject in the undergraduate Bachelors program. The subject focuses on environmental sustainability and engineering safety, drawing on a range of associated case studies. A major assignment for the subject requires students to research and present on a specific safety case study. The research is directed to a specific focus on cause analysis (immediate causes, root causes and contributing factors) as well as both specific technical issues (eg technical engineering aspects and process safety measures) and broader cultural and workplace issues (eg management practices around safety). Students also present impact assessments and regulatory frameworks (including prior requirements, changes following the incident, current requirements and issues of compliance/non-compliance within the case study). Finally students will draw out lessons learned, recommendations following the incident and conduct a critical analysis (for example comparison to similar incidents). Assignment sheets, rubrics and additional resources are provided as supporting material.

2.1. Structure of the case study

The assignment design, with opportunities for complimentary AI activities is outlined below and in Fig. 1:

- The assignment design begins by finding appropriate industry consultants who are available to contribute to the class. We ask these industry consultants to select one case study they feel most qualified and interested to present on. In 2023 this included:
 - o An Air Separation Unit explosion at the Shell middle distillate facility in Bintulu, Malaysia on the 25th December 1997 (Van Hardeveld et al., 2001). Co-author, John Franklin, was a process engineer and manager working at Shell MDS on the day of the explosion, and was involved in the following investigation.
 - o Piper Alpha oil platform explosion on the 6th July 1988 near Aberdeen Scotland (Miller, 1991; Paté-Cornell, 1993; Hull et al., 2002; Shallcross, 2013a, 2013b). Subject consultant Dr Nick Hardy, a Process Safety Engineer, whose consultancy provided expert witness testimony during the Cullen Inquiry investigation following the Piper Alpha disaster, and has also been involved in the implementation of the Inquiry’s recommendations following it.
- Each industry consultant presents a guest lecture to provide background context to their safety case study to the whole class.
- Students are divided into small teams of either 3 or 4 students and given the ability to select a preferred case study (or no preference) for their assignment. For us in 2023, the class divided almost evenly between the 2 case studies, with a small number selecting ‘no preference’ meaning everyone could be allotted to a stated preference.
- Student teams complete background research on the selected case study for approximately 2 weeks.
- Student teams schedule a time with the nominated industry consultant for a 30 minute meeting (within a central timetabling tool, we use Google Sheets)
 - o Complimentary to the in-person meeting with the consultant, AI-consultant chatbots were concurrently released to students for

voluntary access for additional information or to practice interview style questions for the consultation.

- Students write up a report on the incident and submit (the report is assessed, but could be discarded from the activity).
- An AI-academic chatbot was released to the class. This is a chatbot given topic prompts and sample questions related to the themes asked of students in the Viva/defence meeting. This is designed to give student teams the opportunity to practice before the ‘engineering meeting’, for the technical content, meeting conventions and expectations, and also as a forum for the team to practice their interpersonal dynamic (and facilitate intra-team learning among the members).
- Students attend the ‘engineering meeting’ where as a group they are asked prompt questions about the safety case study. All students in the team are required to speak and there is no sub-division of topics (any students can be asked on any topic).

2.2. Customized AI-Chatbot

The AI-Chatbots were not created using OpenAI’s GPTs, which as of the time of implementing these initiatives, did not offer sufficient flexibility of use. Instead it was internally developed to provide greater flexibility. API calls were made to GPT-4 and we used Streamlit as the front end user interface for a text-to-text interaction with the chatbot. Streamlit was useful as a front-end because it offered a professional interface and also a simple URL that could be linked to, within the LMS (making implementation easy and still professional).

We asked the industry consultants to complete detailed questionnaires about their experience and background, and we asked them to answer in their normal speaking style and adopt industry vernacular where possible. The questionnaire itself was only about 2 pages and included questions about their specific background, specific incident details, lessons learned and then perhaps most importantly, their reflections and personal insights. This then informed the system prompt, to provide both the technical details and background of the consultant, but also to allow us to mirror their speaking style (the chatbot can be prompted to reply in a similar tone and adopt common industry language). Additional resources (such as technical papers or news articles) could be added through retrieval augmented generation (RAG).

A critical question, raised within the introduction, is how to deal with ‘hallucinations’. We have placed system prompts and IF statements into the chatbot design, so that it will reply with ‘I’m not sure’ or ‘That is outside of the scope of this activity’ or ask for clarification to questions, when given prompts outside of a channel of replies for the learning activity (as a real person would).

However, a second important feature of the learning design, is the concept of *designed fallibility*. In an industry context, it is essential to foster a workplace culture of shared responsibility for safety; a graduate engineer should not behave in a way they believe to be unsafe, simply because they have been instructed by a more senior colleague. The obligation to create a safe environment, and to perpetually review and cross-check for workplace safety, is shared by everyone, and is always open to question by anyone (regardless of seniority). It is important that this theme is designed into tools and activities intended to educate students about safety: the human consultant advice (and by extension the AI-consultant, mirroring the human roleplay) is meant to guide student thinking, but should not over-determine students’ capacity for independent rational thought, particularly around safety. There is consequently an important but subtle distinction to make: if it were possible to make an infallible chatbot, this would not be an optimal design. This is not to say that errors don’t matter, but rather it is better if the chatbot makes errors in a similar frequency and architecture to the humans it is trying to emulate. It is also important that students have the *perception* that the AI-consultant is open to reproach, to mirror the real-world context it is designed to simulate, so students are still trained to critically evaluate information they are presented. This ‘designed

fallibility' can therefore also accommodate issues of hallucinations, provided they are within the architecture and frequency of normal human error.

After the chatbot was built, we then alpha tested the reliability and validity of the responses from the chatbots among the industry consultants, and the teaching team for the subject (tutors and the subject coordinator, as well as other academics). The chatbots were then made available to students, but with an avenue for students to report errors (ie a normal beta testing approach for testing functionality and performance with real final users). Use was completely voluntary. It was also emphasized that the tool was experimental and may encounter errors. During this beta testing phase, the student interactions were initially monitored directly to look for any deviation from their intended functionality. This was simply an informal process of inviting a few students to use the chatbots early, and directly observing them while using it for the first time (what type of questions do they input, do they use the chatbot as intended). These tests were very positive (anecdotally, students responded well to the chatbot and used it as intended) possibly because the chatbot interface was familiar and intuitive, and directly mirrored a real-world experience (Honig et al., in review).

3. Learning impact of the case study design

3.1. Evaluation design

Following the conclusion of the assignment, we undertook an evaluative study to determine the educational efficacy of the activity, to identify opportunities for improvement and to map the intended learning outcomes against student experiences. As an evaluation of teaching, the primary research question is to address the learning outcomes, with a specific focus on learning and engagement in process safety. These learning outcomes are distributed across the 3 key innovations employed in the new learning design: 1. the use of the experienced industry consultants 2. the use of the longform Viva assessment and 3. the key question of the use of AI-enhancements to (1) and (2).

As a consequence, our research questions were: Was the activity a useful learning experience? This question is further defined within the 3 sub-domains of:

1. Do students perceive consultations with experienced engineers to be useful for learning about chemical engineering safety?
2. Do students perceive vivas/defences to be useful assessment structures for learning about chemical engineering safety?
3. Do our generative AI tools, used to simulate the consultation and viva/defence, improve student learning?

A 'useful' learning experience can encompass numerous pedagogical constructs. For the purpose of evaluation in our study, we frame the utility of the learning activity through 3 key constructs: *perceived* utility as determined by student participants; cognitive engagement meaning the degree of intellectual participation within the activity; and finally the alignment to the intended learning outcomes (the activity may offer broad education value, but we seek to map it to specific outcomes as well). The key intended learning outcomes for the assignment included: Improving safety awareness, deepening understanding of the process safety incident itself, offering practical learning (in the form of authentic assessment) and developing collaborative working skills. The study is therefore focused on evaluation of the learning activity itself and topics related to the assessment (such as the precision of the assessment methodology) are not within the scope of the study.

Given the exploratory nature of the evaluative study, we selected a mixed method research approach. This was preferred because it allows for both quantitative survey design questions, but also direct observation and qualitative feedback, to identify possible limitations not specifically addressed by the prescriptive quantitative survey instrument. Within this mixed methods approach, individual students were first

engaged in informal discussion about the assignment and the dimensions targeted for evaluation. On the basis of these initial exploratory discussions, a survey instrument was developed including both qualitative and quantitative question formats, designed to generate data that could be used to answer the initial research questions. A survey tool was deemed appropriate, not just because it aligns to previous studies (Johns, 2010; Joshi et al., 2015) (thus allowing standardization and comparability to other work) but also because it provides wider access to the target group: it is more accessible (for example to students who may find it difficult to meet in person) and supports anonymous responses, which can lead to more candid and honest feedback. A survey instrument also allows for the use of prompt statement templates from prior validated surveys, which helps validate the current study. We also undertook a reflective study of the outcomes, in line with education design research methodologies (McKenney and Reeves, 2014).

The first author of the paper (Christopher Honig) was the subject coordinator for the subject and as such, may have influenced perspectives on the effectiveness and utility of the teaching tools.

3.2. Survey validation

Before distributing any survey instruments, it is essential to ensure they have first been validated. Validation helps ensure internal consistency and reliability (ensuring everyone interprets the questions as intended) and improves the reliability over time. It can also help provide confidence in test-retest reliability, so respondents reply the same way when question formats are inverted or at different times. To develop an appropriately validated survey instrument, we began by searching for similar studies with validated surveys (Shallcross, 2013a, 2013b; Ocampo-López et al., 2023). To appropriate any prompt statement templates, we need to ensure the surveys were in the public domain. We then looked for studies that were similar to ours and showed strong signs of reliability and validity of their survey tools. While offering a range of directly transferrable Likert templates, the specific goals of our study warranted bespoke prompt statements as well. We therefore created several new custom prompt statements, to elicit student feedback. We then reviewed the complete set of survey prompt statements, for thoroughness and relevance, to make sure it covered all the study's constructs (ie the new teaching and assessment format as well as the use of GenAI). The survey then underwent review by several colleagues, with expertise in engineering education and survey validation, as well as other academic researchers. This process of review helps cross-check the survey for diverse cultural and linguistic suitability, to make sure people from diverse backgrounds still interpret the prompt statements in equivalent ways. This process led to a number of adjustments in phrasing and prompt statements in the survey instrument. Finally we conducted pilot testing of the survey with sample student respondents (these results were excluded from the current study). This pilot testing involved observing students complete the survey and collecting feedback on how they interpreted the phrasing. This created additional confidence in the reliability of the instrument.

It is important to note a common hazard in using surveys designed for teaching evaluation: Student's self-assessed competency is not always a reliable measure of students' actual competency. In aggregate students have been observed to over-estimate their self-assessed competency when developing new skills (McCourt Larres, Ballantine et al., 2003). Student surveys are therefore appropriate for collecting data about the activity (student perceptions, student reference) that can inform conclusions about learning outcomes, but they are not appropriate as a direct measure of student learning (just because a student says they met intended learning outcomes, that does not prove they actually did). It is therefore important to note that in surveying students about their learning, we are in fact measuring student *perception* of their learning and using this as a proxy for their actual learning. For example, it would be methodologically unsound to simply ask students if the learning activity was useful and use this result alone to demonstrate

learning utility. Students may perceive learning that did not take place, they may have power-relationships with the academic, they may falsely self-estimate their competency. Direct student surveying is still a widely used educational research method (Cohen et al., 2002) and is valid provided that the researcher is conscious of the potential pitfalls and takes appropriate steps to control for them. In our research, we have asked questions designed to create data to inform the research question (we have not simply tried to present the final research question or a simple single question validation of the classroom intervention). We have also asked repeated questions, often inverted and distributed the survey after conclusion of the subject (to mitigate any perceived power relationship embedded within the students' obligation to complete the survey). We also note that we have not only used Likert results to draw conclusions. The Likert results are used to support findings from qualitative data and also teacher reflections, in line with education design research methods (McKenney and Reeves, 2014). The alignment of these results is elaborated on further in the discussion of impact section.

3.3. Survey distribution and data handling

The survey instrument was distributed to students via email, requesting voluntary participation. Among a class of 70 students, 52 responded, but only 41 were complete; this offered an effective response rate of 59 %, giving confidence the sample is representative. While this represents a significant sample size of the cohort itself, we note that the small total population size may lower the precision of the findings.

Ethical considerations were paramount for the study, and so in line with the University's ethics approval, the survey began with a plain language statement. This was followed by collection of consent to participate in the study (including pathways for later removal from the study) so students were aware of their role within the study.

The data was collected anonymously and no identifying information was collected except for the IP address of the survey respondent (to ensure no duplicate responses from the same device). After verifying no duplicate responses, the IP addresses were removed from further analysis so the data was fully deidentified.

We note that in beta-testing with students (discussed in Section 2.2) that student identities were known to academic researchers; we were directly observing their use of the chatbots. These students were not sent the survey link, and so their responses do not appear in the survey results.

3.4. Quantitative data analysis

For the quantitative component of the survey tool, responses were gathered using the universally recognized 5-point Likert scale to judge self-perceived alignment to a series of prompt statements. This offers greater translation to other work, because of its standardized and widely adopted structure (Johns, 2010; Joshi et al., 2015). The response options were formatted to the standard 5 statement options: "strongly disagree"; "disagree"; "neither agree nor disagree"; "agree"; "strongly agree". Where appropriate, a 6th option was made available formatted as "not appropriate/did not use". These statements were converted to numerical scores (1: "strongly disagree"; 5: "strongly agree", "did not use" was removed from the study). Descriptive statistics were used to summarize the data for analysis (mean, standard deviation) but have been presented in the paper as averages and percentage of respondents who agree/disagree, to make the findings clearer and easier to interpret. The respondent percentages who 'agree' with a prompt statement include Likert responses of both 'agree' and 'strongly agree' (scores above 3) and similarly respondent percentages who 'disagree' include both 'disagree' and 'strongly disagree' statements within Likert survey (scores below 3).

3.5. Qualitative data analysis

For the qualitative component of the survey, we included questions

designed to gather feedback that could provide data to address the research question. While various methodologies are available for processing qualitative data, we opted for content analysis (Cohen et al., 2002; Krippendorff, 2018). This method was preferred as it allows for an in-depth examination of student responses. Although not a deciding factor, the manageable data size ($n = 41$) made this method feasible.

Content analysis is versatile and can be adapted across different theoretical frameworks, making it widely used and familiar to a broad audience (Cohen et al., 2002). Its generalizability to related studies further supports its use. However, a well-known limitation of content analysis is its susceptibility to subjectivity and bias due to the interpretative nature of responses. Despite this, content analysis is often favoured for giving a 'voice to the dataset,' which we consider beneficial for the exploratory nature of our research, even though it may limit the theoretical rigor.

There are numerous bespoke approaches and subtle modifications to content analysis methodology (Cohen et al., 2002), but our approach mostly closely followed Krippendorff's conventions (Krippendorff, 2018). We also note that the content analysis methodology overlaps significantly with thematic analysis (Thomas and Harden, 2008; Vaismoradi et al., 2016; Clarke and Braun, 2017; Castleberry and Nolen, 2018), as both are rooted in the interpretivist paradigm, and both may be suitable for our dataset (Cohen et al., 2002).

We then used an inductive coding process, allowing a bottom-up interpretation of the data that allows codes and themes to be derived from the data itself. Inductive coding favours the exploratory nature of the study, by allowing patterns, themes and insights to emerge, that are not pre-determined. The advantages of this approach is that it's grounded in the data itself, so it can capture unexpected nuances, but inductive coding also lacks a clearly pre-defined framework and can be biased by the researchers' subjective perspectives.

We began by thoroughly reading all responses to understand the overarching sentiments, for data familiarization, before starting the coding process. This initial reading helped us identify recurring sentiments, and specific phrases used by the respondents. Within the inductive coding process, through multiple iterations, we refined our codes, selecting individual phrases from student responses, that captured the underlying sentiment, and condensed these between responses (Krippendorff, 2018). Initially this involved annotating the data and highlighting significant phrases or terms. We then grouped similar codes to form broader categories and developed themes that grouped several ideas together. We then reviewed the themes for coherence and internal consistency with the data itself. Sample codes and identified themes are shown with the data in Section 4. The data was recoded 4 months later using the original codes, to confirm intra-coder reliability over time, from a single person. We found a high reproducibility over time (all coding matched except for 2 code mappings). However, we did not use a second person to code the data to confirm inter-coder reliability (Krippendorff's Alpha).

The small dataset size facilitated the construction of a comprehensive code library, allowing us to perform all coding in Excel without specialized software like NVivo. We developed an extensive library of content codes ($n = 33$), which we then grouped under common themes identified within the data; individual codes could often be mapped to a single theme. While some themes were related and could be further combined, we prefer to leave them in more general forms for methodological transparency.

4. Discussion of impact

The study is an evaluation of the impact of the learning design, but given the use of 3 key innovations simultaneously, that are difficult to decouple from overall learning outcomes, we have focused on these 3 domains specifically: 1) Use of experienced highly experienced (semi-retiree engineers) as industry consultants 2) The use of long-form oral assessment in the 'engineering meeting' and 3) the use of generative AI

to enhance items (1) and (2).

4.1. Student response to industry consultants

We identified a strong positive student response to the inclusion of the industry consultants within the assessment design equal across both consultants (Likert Question 4.1.1, Table 1). This was also reflected within the qualitative results (Tables 2 and 3). When asked ‘what aspects of this assignment did you find most beneficial to your learning’, the thematic of ‘access to industry consultants’ was the most prevalent identified (15 individual responses, or 37 % of respondents) and also in the inverted form, when asked ‘What improvements would you suggest to this assignment?’, it was the second most common thematic, essentially asking for ‘more time with consultants’ through either the code of ‘longer meetings’ or ‘additional meetings’ (7 individual responses, or 17 % of respondents).

However, the result was more muted when we controlled for simply offering more direct access to teaching staff (Question 4.1.2, Table 1). When framing the statement directly in comparison to more time with the lecturer, only 49 % of respondents agreed and 39 % actively expressed no preference. This indicates a preference for more time with teaching staff, not necessarily more time with industry consultants specifically. Or to frame this another way, there can be perceived value in complimentary roles of both academics and industry consultants, and while students valued time with consultants, they also value direct access teaching staff, such as lecturers, approximately in equal measure.

4.2. Response to ‘engineering meeting’ (viva/defense)

Like the industry consultation, the ‘engineering meeting’ in a viva format also observed strong student affirmation, across self-assessed learning experience, perceived practicality and in comparison to other assessment formats (Questions 4.2.1–4.2.3). We note that the preference for this assessment format over other modes, created more diverse responses, with a heavier weighting indicating no preference (14 responses, or 34 % of respondents indicated no preference, Question 4.2.3) (Table 4).

Similarly, there was positive support observed in the qualitative data. This is consistent with the findings Section 4, given the interactive and collegiate format of the assessment, it functions similar to an additional consultation meeting.

Although not directly assessed within the paper, it is worth noting that traditional report formats are susceptible to ghost writing, particularly with the rise and accessibility of new GPT tools (for example ChatGPT, Grok, Gemini). Oral presentations with a slide deck are also susceptible to GPT-assisted formatting and writing but the in-person nature of a viva, and its absence of pre-prepared written material, mean the student’s knowledge and understanding of the topic can be more completely understood. It is also useful to have interactivity and to be able to query student statements for underlying understanding, in

Table 1

Likert survey results for prompt statements relating to the Industry Consultations. The percentage of respondents who agree, represent the statements “strongly agree” and “agree” in the Likert survey and similarly the percentage of respondents who disagree represents the statements “strongly disagree” and “disagree” within the Likert survey.

Prompt statement	Average	Agree/Disagree percentages
4.1.1 The initial meeting with industry consultants deepened my understanding of the real-world safety case study	4.3	Agree = 80 % Disagree = 7 %
4.1.2 I would prefer more consultation time with industry experts (compared to more consultation time with lecturers in the department).	3.9	Agree = 49 % Disagree = 12 %

Table 2

Processed qualitative results to the prompt question: What aspects of this assignment did you find most beneficial for your learning? Most frequent themes and codes.

Categories	Codes and sample responses	Frequency
Consultation with industry experts	Consultant directly: “The chance to talk to the industry expert in the consultation.” Expertise: “I think that the consultant expertise helped”	15
Improved safety awareness	Safety learning: “Learning about a past safety incident” Safety analysis frameworks: “Learning the types of safety analysis”	13
Practical Experience	Practical setting: “...how these are conducted in a practical setting” Real-world: “...it felt like it was actually preparing us for real world engineering scenarios”	10
Background research	Independent research aspect: “Research and critical thinking around mitigating risk.”	8
Experience of Engineering Meeting	“Consultation and defence are the parts that I found most beneficial for my learning. These parts provides better engagement with the project.”	3
Collaboration with students	“The experience of working on a larger assignment with a group was useful as we had to decide how to break up the work between us, set internal deadlines and organised several meetings”	3

Table 3

Processed qualitative results to the prompt question: what improvements would you suggest to this assignment? Most frequent themes and codes.

Categories	Codes and sample responses	Frequency
Student explicitly replies with no changes necessary	None: “None, I thought it was good”	9
Consultant access	Longer meetings: “...The only thing I could suggest as an improvement from my perspective is that engineering meetings could be a bit longer.” Additional/More frequent meeting: “Potentially having the opportunity to have a two meetings with the consultants as I found these discussions were where a greater understanding was made.”	7
Access to information	Difficulty finding case study information: “...to be honest it was quite difficult to find additional info required...”	7

Table 4

Processed Likert responses related to the engineering meeting activity.

Prompt statement	Average	Agree/Disagree percentages
4.2.1 The defence (engineering meeting) between our team, the industry consultant and lecturer, was a useful learning experience	4.5	Agree = 90 % Disagree = 2 %
4.2.2 The defence (engineering meeting) offered a more practical learning experience than other assessment (exams, assignments)	4.1	Agree = 80 % Disagree = 7 %
4.2.3 I would prefer more assessment like the defence (‘engineering meeting’) compared to other forms of assessment	3.9	Agree = 59 % Disagree = 7 %

assessing underlying comprehension, particularly when the content of final written reports (as a proxy for student comprehension) are increasingly open to accessible ghost-writing tools.

4.3. Response to AI-enhancement

A strong uptake of the use of AI tools was observed, but a more muted student response to the perceived usefulness of our tools. However this result was still encouraging given we conceived of the AI tools as solving issues of scale of teaching and we thought of them as innately inferior to in-person teaching (students perceived them more favourably than we expected). We note that ‘performance expectancy’ of the AI-tool was the primary student concern to adoption, which has been discussed in a separate paper investigating barriers to use of AI teaching tools.

The voluntary uptake of the AI-tools was generally high. Of the 41 respondents:

- 34 reported using the AI-consultant and 7 reported not using it (83 % voluntary uptake among respondents).
- 29 reported using the AI-‘engineering meeting’ and 12 reported not using it (71 % voluntary uptake among respondents).

The high self-reported use could also be verified against actual use of the platform. We recorded the number of unique IP addresses accessing the Streamlit hosting website (but not the actual IP addresses themselves). We recorded 78 unique users (in a class of 70) but 4 of these corresponded to teaching staff. Students were able to access the platform across multiple times with different devices, accounting for a final number higher than the total number of students. This result is indicative of a high rate of student uptake and helps verify the self-reported use.

Student responses (Likert prompt statements 4.3.1–4.3.3, Table 1) were broadly positive, for example 58 % of respondents rated it as a useful learning experience. This positive response was a surprise because we initially conceived the AI-consultant and AI-‘engineering meeting’ as solving the issues budget and workload of the teaching staff. This result was interesting: the AI tools provided not only increased teaching volume, but additional value in of themselves.

In initial student interviews to develop the survey instrument, we identified a sub-group of students who expressed a preference for the AI-tools. When asked why, these students expressed concern about ‘losing face’ when asking questions in a group setting. This prompted the inclusion of question 4.3.3, to gauge student preference for the consultant based- 12 respondents (29 % of respondents) said they felt more comfort speaking to the AI-consultant (‘agree’ or ‘strongly agree’). We surmise this is related to a students’ sense of introversion, but note this only a hypothesis, not a rigorous finding of the paper. It was an unexpected result but suggests other possible strategic use-cases of AI-teaching tools, particularly to help students who may have social or performance anxieties in other assessment formats (Tables 5–9).

In support of the quantitative result, within the qualitative data, there was again a high frequency of responses indicating the AI tool was useful (12), compared against the frequency of users who explicitly stated some aspect was not useful (8).

Among the themes correlated to positive response and drivers for use were primarily that it could be used as a research tool, either at the end to verify information (6) or during the initial research phase (5). There

Table 5

Processed Likert responses related to the AI-consultant.

Prompt statement	Average	Agree/Disagree percentages
4.3.1 The AI-consultant Chatbot offered a useful learning experience	3.8	Agree = 59 % Disagree = 10 %
4.3.2 Practicing with the AI-Lecturer Chatbot before the defense (engineering meeting) helped me feel more prepared.	4.1	Agree = 61 % Disagree = 7 %
4.3.3 I feel more comfortable asking an AI-Consultant chatbot questions (compared to a real consultant).	2.8	Agree = 29 % Disagree = 49 %

Table 6

Processed qualitative results to the prompt question: did you make use of the AI consultant chatbot? If not please explain why. If you did, please outline how useful you found it.

Categories	Codes and sample responses	Pos/ Neg	Frequency
Was Useful	“Yes, I found it useful in terms of preparation for the final meeting and to verify any minor information that might have passed my head regarding the case.”	Pos	12
Wasn’t Useful	“...the feedbacks weren’t really useful...”	Neg	8
Review or verify information	“I found it useful ... to verify any minor information that might have passed my head regarding the case”	Pos	6
Limited scope	“... it became fairly repetitive after interacting several times.”	Neg	6
Prefer human consultant	“I felt it was more useful to ... consult the actual expert.”	Neg	5
Inaccuracy or misinformation concerns	“Felt that use of the chatbot was risky as it might make me learn false details of the case study”	Neg	5
Used for research	Preliminary research “Yes, I used it to figure out small details that preliminary research” Information reaffirmation “Yes, I used the Chatbot to reaffirm my knowledge of the Bintulu explosion”	Pos	5
Discomfort with AI	“I did not make use of the chatbot because I am generally uncomfortable and mistrustful of AI”	Neg	3
AI is clearer (than online sources)	“I also think that the AI consultant responded to my questions in a more simplistic and clear manner in comparison to many of the reports/articles I found online.”	Pos	3
Prefer self-research	“...I felt it was more useful to either perform my own research or consult the actual expert.”	Neg	2
Speed of replies	“...the chatbot gave a quick response and improved my learning efficiency”	Pos	2
Insufficient time	“...by the time it was available I was more focused on finishing the report and prepping myself for the defence and I don’t think I could really gain a lot from it.”	Neg	2
Questions after consultation	“It also served a great way to ask follow up questions that had come up after consultation with the consultant”	Pos	2
AI comprehension of question	“It was sometimes hard for the AI chatbot to understand the exact question asked...”	Neg	1
Helps language barriers	“I find it helpful especially to students who are struggling on understanding the teachers accent or English”	Pos	1

were also several themes related to preference for the AI for improved communication: because it is clearer and can offer non-technical explanations (3), for overcoming language barriers (1) and for its speed of replies (2).

The themes identified that were mapped to negative views of the tool, and may function as barriers to uptake, were much more varied. Several expressed concerns with the reliability of the tool itself, either that it may give misinformation (5), that its scope of information was limited (6) or that it misinterpreted question (1). Beyond issues related directly to the tool itself, many respondent themes described a relationship to GenAI, and personal preferences, unrelated to the design of the tool. These included an innate discomfort with AI (3) or preferences

Table 7

Processed qualitative results to the prompt question: did you make use of the AI-Lecturer Chatbot? If not please explain why, if different from above. If you did, please outline how useful you found it.

Categories	Codes and sample responses	Pos/ Neg	Frequency
Was Useful	"Yes, it was an opportunity to summarize what we have researched, put them in order and present it to someone (even AI)"	Pos	9
Wasn't Useful	"I didn't feel like I need to use it (not significantly more informative than regular chatgpt)"	Neg	3
Guide to thoroughness/expectations	Practice "Yes I used the chatbot. Primarily for deducing the style and type of questions which might be asked in the defence"	Pos	16
Preparation for engineering meeting	Feeling more prepared "...This helped me to feel more prepared before my first defence in ChemE." Used to prepare "...I found this very useful in preparing for the engineering meeting..."	Pos	7
Insufficient time	"I didn't use it mainly because I didn't give myself enough time to use it"	Neg	6
Inaccuracy Concern	"I did not use this AI tool because I think the response it generated will deviate significantly from the real world scenario."	Neg	2
Misinformation risk	"The first time I used it gave me misleading information and it sent me into a panic of being unsure of the information I already knew"	Neg	2
Lack Detail	"...some detailed informations were not revealed. We need to consult the literature to know them."	Neg	2
Prefer direct practice with peers	"...I have a method of practising with my team mates so we did it worth ourselves and also brainstormed on what type of questions might be asked."	Neg	2

Table 8

Processed Likert responses related to student comfort and stress during the activities.

Prompt statement	Average	Agree/Disagree percentages
4.4.1 I felt comfortable asking questions during the consultation.	4.3	0.7
4.4.2 felt comfortable speaking during the defense (engineering meeting)	4.3	0.8
4.4.3 The defence (engineering meeting) was more stressful than other assessment (exams, assignments)	2.7	1.2

Table 9

Likert responses to general student feedback related to the activity.

Prompt statement	Average	Agree/Disagree percentages
4.5.1 The assignment was a useful learning experience	4.65	Agree = 100 % Disagree = 0 %
4.5.2 This assignment has increased my interest in safety and risk management in chemical engineering.	4.4	Agree = 85 % Disagree = 2.5 %

for familiar methods of study, such as self-research online (2) or human consultation (5).

4.4. Student stress

During the assessment process, in direct observation of students within the consultation and the 'engineering meeting' (viva), a number of students appeared highly anxious, despite the best efforts of the assessors to make it relaxed and informal environment. This was to the point that some students effectively 'froze'; they struggled to think on their feet and formulate clear answers to questions, meaning their anxiousness could interfere with the assessment of their underlying learning.

To this point, we included questions to assess students' self-perceived stress during the assessment items, and comparative questions to other assessment formats (prompt statements 4.4.1–4.4.3) (Joughin, 2010; Ringeisen et al., 2019). Despite the direct observation that some students appeared very anxious within the meeting, this wasn't reflected in the retrospective student self-assessment in the aggregated data. Most students expressed comfort in speaking during the consultations and defence, and did not rate it more stressful than other assessment items. In the qualitative data, no students mentioned a stressful environment within the interpersonal meetings (it did not appear thematically in any student response). Our observations are only anecdotal and we do not really have a clear explanation for this, other than to propose 2 possible theories:

1. Retrospective bias. The survey instrument only assesses the student's perception of the experience after the event, so it measures the recollection of the experience, rather than the direct experience, in the moment. Although the assessment may be stressful in the moment, the retrospective recollection of this stress is minimized (ie the narrative 'it wasn't that bad in the end').
2. Comparative stress and visibility. The assessment was not significantly more stressful than other assessment formats, but the interpersonal meetings just made this more apparent to the teaching staff (ie the staff could directly observe how stressful assessment can be for some students, prompting the line of inquiry, but exams and written assignment can be comparably stressful) (Joughin, 2010).

4.5. Overall learning experience

A key question is how well the teaching design delivered the intended learning outcomes to students. It's important to note that this cannot be reliably determined by just asking students in a Likert survey or qualitative questionnaire: these question formats only assess student *perception* of their learning outcomes. It is well understood that when assessing self-perceived learning outcomes, respondent bias typically over-estimates self-assessed competence (McCourt Larres et al., 2003) but these types of questions are still useful for understanding students' perceived value of the activity, as a proxy for student intellectual engagement. With these distinctions and limitations in mind, we found students strongly perceived the activity to offer a useful learning experience. All respondents agreed with the statement "Take home assignment 2 was a useful learning experience" (Question 4.5.1) with a Likert response of 4.65 indicating very strong student endorsement of the activity. The result was also reproduced in the qualitative data in an inverted question framing ("What improvements would you suggest to this assignment?"; Table 3) the most frequent thematic response was for no change to the assignment design ("none", "no change", reaffirmation of the design). All these results indicate strong student *belief* in the value of the assignment design for learning, and consequently stronger student engagement in the activity and strong endorsement for the assessment design. We note that while we had a significant representative sample of the cohort of students (59 % response rate) the number of respondents was limited by the overall class size (N = 41, 70 total students within the

class) which may limit the precision of the result. For this specific result however, we note the extremely high concordance in rating gives strong evidence to support the learning design: ALL 41/41 respondents agreed the learning activity was useful, which evidences the sample size has not been an issue in drawing the conclusion that students perceived value in the learning activity.

A key intended learning outcome for the assignment is to engage students in safety analysis and process safety. Doing this in an engaging and constructive way can increase student interest in the field, and so this learning outcome can be assessed by collecting data on student awareness and interest in pursuing safety and risk management in chemical engineering, following completion of the assignment. This is reflected in Question 4.5.2 which evidenced 85 % agreement (Likert score 4.4) to the statement ‘This assignment has increased my interest in safety and risk management in chemical engineering’. A strong endorsement for the assessment design.

Finally a key question of the paper is whether the learnings demonstrated by the students were significantly improved through these 3 major changes. The assessment in both years (before and after the changes) was conducted by 2 examiners (the subject coordinator and an industry consultant) through the application of a grading rubric. This rubric assesses students across 3 domains: 1. critical thinking (student ability to offer new insights), 2. technical understanding (factual accuracy in understanding the events) and 3. structure (clarity of explanations and presentation within the meeting). Within each category, we have grading bands with up to a dozen examples and descriptors of presentation styles that would be situated within this range. This helps mitigate subjective judgement of student performance. Although we acknowledge that there may be researcher biases with this grading system, we feel it is valid to comment directly on aggregated student performance between the years. Here we directly observed that students displayed significantly better comprehension of the safety case study, in comparison to previous years, consistent with the survey findings. When applying essentially the same rubric to the activities over the 2 years, the aggregated student result went from 77.0 to 86.3, using the same examiner. Assuming the 2 cohorts are of equal capability (at the same year level, having completed the same pre-requisites) the only other difference is the intervention with the new classroom activity, and so we attribute this improvement to the intervention. Beyond the specific marks awarded, from the perspective of Education Design Research (McKenney and Reeves, 2014) we note that within teaching reflections, we observed a marked improvement in student learning, aligned to the intended learning outcomes, of improved critical thinking and deeper engagement with the learnings of the safety case study. From these observations, we believe these 3 modifications offer significant enhance to safety education.

4.6. GPT cost implications and API tracking

The discussion of new teaching initiatives must be balanced against the resourcing required to implement them. For example personalized one-on-one tutoring might offer better learning outcomes for students, but this may not be feasible within constrained operating budgets.

At the time of running the study, OpenAI did not offer the ability to track token count or API usage within the user dashboard, but did offer daily billing data. When tracking the API-key for our GPT-tools, we were able to calculate the total token spend (for the full class for all activities contained within this paper) at \$USD 34.74 (the sum of the daily billing over all days the chatbot was live). By counting the tokens within the system prompt and estimating that an average student query contains 200 tokens and an average reply contains 500 tokens, we were able to estimate the total number of API requests at just over 1000. This averages to about 15 question-answers per student within the class (ie 15 question and answers from the GPT per enrolled student). The key implication of this finding is firstly that the tool was widely utilized, consistent with the findings in Section 4.3. More importantly though, the

cost of implementation of using the tool was insignificant; this learning activity cost less than 50 cents per student (USD). GPT teaching tools offer huge advantages for education: scalability, universal access, instantaneous replies but they also do it at significantly lower costs. Our study finds they also offer substantive learning outcomes, but even if they only match current teaching practices, these other cost and implementation benefits mean they will have significant impact on the educational landscape.

4.7. Improvements

In completing the study, we identified a range of improvements for future iterations, but these primarily relate to the design of generative AI tools. Going forward we intend to structure the system prompt to create shorter replies (to reduce token costs and also limit the information provided in each reply). We are also investigating voice-to-text plugins and avatars to enhance the verisimilitude of the activity.

GPTs offer a range of wholly new activities for students. Where learning is often conducted passively when studying alone (for example reading a textbook, completing tutorial problems) the key advantage of GPT-enabled teaching tools is their ability to enable new interactive learning activities, aligning to Constructivist Learning Design principles.

In a seemingly oxymoronic finding, the chatbot was found to be both too helpful and not helpful enough. For example, although we tried to design the AI Consultant to ‘drip feed’ information to students (requiring continuous engagement to ask ongoing questions) it often reverted to its underlying ChatGPT ‘nature’ of trying to be accommodating and helpful, and would often provide excessive information (in a way that is unlike a human in its roleplay function, and so breaks the verisimilitude of the roleplay). Then simultaneously, as can be seen in the student written feedback, it was commonly perceived to provide insufficient detail or offer equivalent information, that students already found in their research. We believe these issues can be resolved by iterating the system prompt design and including retrieval augmented generation with additional technical resources.

5. Conclusion

Effective process safety education in chemical engineering and process engineering is essential if we are to prevent and/or mitigate future disasters, improve process efficiency and protect personnel. We offer three suggestions to enhance learning outcomes for students within safety education:

1. Drawing on the experience of senior professional and consulting engineers, approaching retirement, to transfer intellectual capital to students, through consultations, particularly in investigating safety case-studies aligned to the experience of the consultant engineer.
2. The use of long-form interactive assessment through a viva, simulated as an ‘engineering meeting’ to drive the appropriate learning practices. These types of assessment formats are particularly important to assess and verify students’ underlying understanding, given the emergent accessibility of GPT ghost-writing. Longform viva assessment also creates opportunities to meaningfully continue the learning activity, within the assessment format.
3. The use of GPT-powered tools to enhance items (1) and (2), to provide greater scale and access for students as well as improved engagement. We also note the use of GPT teaching tools can improve access to learning, particularly for students who may be introverted; some students express a preference for these tools.

We evaluated the teaching practice through a mixed methods approach and direct observation, to show student preference for all 3 items, student *perception* of improved learning outcomes and directly observed improvement.

CRediT authorship contribution statement

Christopher David Frederick Honig: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Aditya Desu:** Software. **John Franklin:** Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT-4 in order to improve writing layouts and suggest writing formats to improve readability and clarity of the underlying ideas. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of interest

The authors declare no competing interests.

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